Drug Search and Physician Hazard: An Investigation into Addict Behavior and Policy Remedies

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Roadmap

Roadmap

Overview of Paper

- The prescription drug epidemic is a major public health crisis in the U.S.
- National Institute on Drug Abuse (NIDA) reports over 25,000 deaths from prescription drugs in 2023[1]
- This paper investigates the relationship between addict drug search and physician prescribing behavior
- **Key Question: How do physician prescribing patterns and addict drug search behavior interact and change in response to policy interventions?**

[1] https://nida.nih.gov/research-topics/trends-statistics/overdose-death-ratesFig2

Contribution (Short Discussion)

1. **Addiction Literature:** This paper contributes to the addiction literature by pivoting from economic loss analysis or treatment efficacy to examining interactions between addict search behavior and physician prescribing patterns via dynamic programming.
2. **Drug Abuse Literature:** This paper adds to the literature surrounding drug abuse by looking at behavioral patterns and responses to policy changes, providing insights into how addicts adapt their search strategies in response to changes in physician behavior.
3. **Policy Implications:** The findings of this paper have implications for public health policy, particularly in motivating interventions that effectively address both physician prescribing practices while also ensuring addicts are not unduly harmed.

Roadmap

1. Paper Overview & Contribution
2. Story & Key Findings
3. Ideal Data
4. Empirical Strategy
5. Results
6. Conclusion and Next Steps

Key Findings

- **Finding 1:** Addicts are at the whim of prescribers
- **Finding 2:** Policy interventions targeting physician prescribing practices can effectively reduce the availability of prescription drugs, but may also lead to unintended consequences in addict search behavior.
- **Finding 3:** Dynamic programming models reveal that addicts adapt their search strategies based on perceived changes in drug availability and physician behavior.

Roadmap

Data Needs

- **Addict Behavior Data:** Modeling the Future provides panel data which would be useful in tracking addict's tolerance evolution over time, as well as their constraints and exit conditions from the model.
- **Claims Data:** Prescription claims data would grant me insight into both prescribing behavior and search behavior, allowing me to see how addicts interact with physicians and how prescribing patterns change over time.
- **Physician Data:** Data from the AMA or state medical boards would provide information on how risk tolerance of physicians evolves, how the distribution of physician types looks, and to understand the exit conditions for physicians

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Model - Addicts

- $\exists$ a continuum of addicts $\mathcal{A} = \{a_1, a_2, \ldots, a_N\}$
- Each addict a_i has a tolerance endowment, γ_i, which takes one of two values: γ_L or γ_H
- Each period of search, addicts have a probability θ of matching with a prescribing physician
- Key variables include:
  - s: search cost
  - r_i: risk tolerance of addict i
  - β: discount factor
  - $p_i t$: amount prescribed to addict i at time t
  - ε_a: shock to addict's risk tolerance

Addict's Problem

In the first period:

$$V_a^0(\gamma_i, r_i) = u(c_i) + \beta(\mathbb{E}(1-\theta)V_a(\gamma_i, r_i) + \mathbb{E}\theta V_a(\gamma, r'))$$

s.t.

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$$

$$r' = r_i + \varepsilon_a$$

Addict's Problem (Cont.)

In subsequent periods:

$$V_a^m(\gamma_i, r_i) = \max\{V_a^s(\gamma_i, r_i), V_a^f(\gamma_i, r_i')\}$$

s.t.

$$V_a^s(\gamma_i, r_i) = \{u(c_i) + \beta V_a^m(\gamma_i, r_i')\}$$

$$V_a^f(\gamma_i, r_i') = \{u(c_i) + \mathbb{E}_t[\beta(1-\theta)V_a(\gamma_i, r_i') + \beta\mathbb{E}_t[\theta V_a(\gamma_i, r_i')]]\}$$

$$u(c) = \frac{c_i^{1-\sigma}}{1-\sigma}$$

$$r_i' = r_i + \epsilon_a$$

$$c_i = \gamma_i p_{it} - r_i^{1/\gamma_i} \times \sum s_{it}$$

Model - Physicians

- There is a continuum of physicicans $\mathcal{M} = \{m_1, m_2, ..., m_K\}$
- Each physician m_j has a risk tolerance r_j which takes one of two values: r_L or r_H
- Physicians are profit maximizing agents who choose to prescribe or not prescribe to addicts based on their risk tolerance and the potential profits from prescribing
- There is a subset $\mathcal{I} \subset \mathcal{A}$ of addicts who visit physician m_j each period, such that each doctor can see many addicts each period

Physician's Problem

$$\max_{v_j} v_j - i_j r_{jt}$$

s.t.

$$v_j = r_m^2 \sum_{i=1}^{I} p_{it} + \sum_{i=1}^{I} a_{it}$$

Key variables include:

- v_j: Revenue from prescribing
- i_j: Cost of prescribing (legal risk, professional risk)
- a: number of addicts seen in period t

Estimation

- Solve dynamic programming problems for addicts and physicians using value function iteration
- For each iteration:
- Solve addict problem
- Solve physician problem given addict distribution
- Update prescribing distribution among physicians
- Check for convergence
- Simulate model and counterfactuals to analyze policy interventions

Simulation Parameters

Set key parameters:

- Discount factor $\beta = 0.95$
- Tolerance levels $\gamma_L = 0.5$, $\gamma_H = 1.5$
- Search cost $s = 1$
- Risk caps: $r_m = \{10, 30\}$, $r_a = \{5, 15\}$
- Probability of prescribing $a(\theta) = 0.3$
- Risk tolerance shocks $\varepsilon_a \sim N(0, 1)$
- Number of addicts $N = 1000$, Number of physicians $K = 600$

Roadmap

Results

Table: Baseline and Counterfactual Summary Statistics for Doctors

Avg. Profit	50.92
Share Exited	49%
Counterfactual Avg. Profit	48.47
Counterfactual Share Exited	75%

Table: Baseline and Counterfactual Summary Statistics for Addicts

Avg. Value	5.80
Avg. Search Stock	5.78
Difference in Value - CF and Baseline	-3.99
Difference in Search Stock - CF and Baseline	3.98

Results (Cont.)

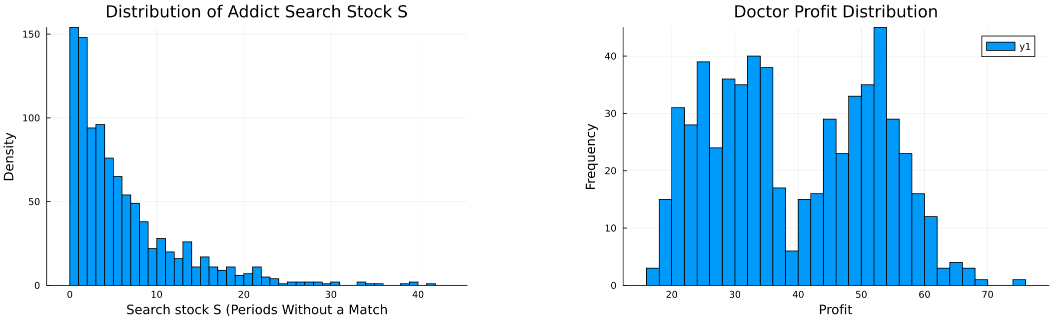
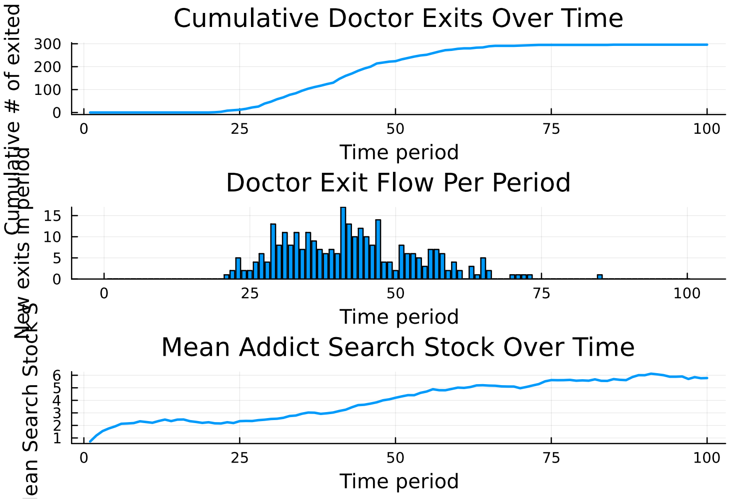


Figure: Distributions of Addict Search Stock and Doctor Profit

Results (Cont.)



Counterfactual Results

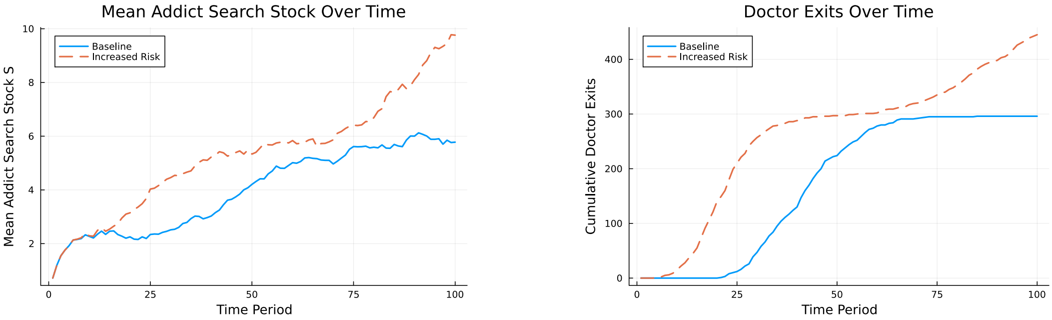


Figure: Counterfactual Distributions of Addict Search Stock and Doctor Profit

Roadmap

Next Steps

- Calibrate to actual data
- Build on basis through dynamic types and learning
- Explore additional policy interventions

Conclusion

- This paper develops a dynamic programming model to analyze the interaction between addict search behavior and physician prescribing patterns.
- Key findings indicate that policy interventions targeting physician behavior can significantly impact addict search strategies, with important implications for public health policy.
- Policy makers should consider the dynamic responses of both addicts and physicians when designing interventions to address prescription drug abuse.
- Future work will focus on calibrating the model to real-world data and exploring additional policy scenarios to inform effective interventions.

Thank You

Questions?